

SOUMABRATA GHOSH

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TECHNICAL SKILLS

Languages: C++20, C, TypeScript, JavaScript, Python, SQL, POSIX Shell/Bash

Distributed Systems: Redis Lua atomic leases, Apache Kafka (partitioning, consumer groups), Distributed Locks (Redlock), WebSockets, Circuit Breakers, Dead-Letter Queues

Databases & Storage: SQLite FTS5 / BM25, Turso libSQL, PostgreSQL (MVCC, Index Optimization), MongoDB, Redis Streams & Pub/Sub

Infrastructure & Tooling: Docker containerization, Linux eBPF/perf, Locust Load Testing, Git, Next.js, CI/CD pipelines

WORK EXPERIENCE

ShapeNXT.com | Full-Stack Software Engineer Intern

Jul 2026 – Present | Remote

- Engineered and shipped production features across company web platform, LMS (learn.shapenxt.com), and internal CRM using Next.js, TypeScript, PostgreSQL, and MongoDB.
- Rebuilt company marketing infrastructure from scratch, owning full-stack implementation, performance tuning, and automated deployment pipelines.
- Architected in-platform blog content authoring and publishing system for the LMS, enabling autonomous course marketing without engineering dependencies.

FEATURED ENGINEERING PROJECTS

InstaRide | C++20, Redis Lua, PR-Quadtree, SIMD AVX-512, Docker, WebSockets

2026

- Engineered a sub-millisecond spatial dispatch kernel in **C++20**, processing **3.88M ops/sec** spatial ingestion with **16.5 µs** p99 k-NN query latency (75x faster than PostGIS baselines).
- Designed a Point-Region Quadtree with Struct-of-Arrays (SoA) layout and vectorized **AVX-512** bounding box pruning evaluating 16 coordinates per clock cycle with zero heap allocations.
- Eliminated distributed double-booking anomalies using single-roundtrip atomic **Redis Lua** scripts handling **71.9k lock acquisitions/sec** with zero race collisions under chaos testing.
- Hardened engine against singularity spatial clustering (50k drivers on 1 coordinate) via depth-clamped bucket overflow; verified 0 segfaults under ASan/Valgrind.

Distributed Full-Text Search Engine | TypeScript, SQLite FTS5, Turso libSQL, BM25 Ranking

2026

- Built an inverted index search core supporting full-text retrieval across 100,000+ documents with sub-**5ms** search response times and tokenized trigram query routing.
- Implemented custom **BM25** relevance scoring with dynamic term frequency saturation and document length normalization for high-precision query evaluation.
- Constructed edge database replica sync using Turso libSQL, reducing p95 query latency by **65%** for distributed multi-region clients.

High-Throughput Webhook Relay Fabric | Apache Kafka, Redis, Circuit Breakers, Node.js

2026

- Architected an at-least-once distributed event ingestion fabric sustaining **120,000 req/sec** ingress with durable stream buffering.
- Engineered partitioned **Apache Kafka** topic pipelines with automated exponential backoff retries and Dead-Letter Queues (DLQ) to isolate failing subscriber endpoints.
- Integrated distributed token-bucket rate limiters in **Redis** with sliding-window accounting to defend downstream webhooks from traffic storms.

Atomic Event Ticket Reservation Engine | PostgreSQL, Redis, Distributed Locking, Node.js

2026

- Designed a high-concurrency seat reservation system sustaining **1,000+ RPS** concurrent burst traffic under Locust stress testing with **0.000%** over-allocation.
- Combined Redis temporary reservation TTL leases with PostgreSQL MVCC and optimistic row locking to guarantee strict transactional consistency under heavy contention.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

Kolkata, India | 2026

Relevant Coursework: Operating Systems, Distributed Systems, Computer Networks, Database Internals, Data Structures & Algorithms, Object-Oriented System Design.

ENGINEERING STRENGTHS & VERIFICATION

- Code Volume & Reliability:** Authored 48,000+ lines of production code across 5 distributed architectures with zero memory safety violations.
- Chaos Testing:** Engineered custom load injection harnesses simulating network partitions, split-brain nodes, and packet drops under continuous Locust profiling.